

IA Applied to IIoT Intrusion Detection: An Overview

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Abstract—Over years, the Industrial Internet of Things (IIoT) has evolved rapidly, offering increased connectivity and benefits in terms of efficiency and productivity as well as significant business opportunities. It is used in various fields such as transportation, production, supply chain management, the oil and gas sector, mining and metallurgy, energy services, aviation, etc. However, this increased connectivity also exposes industrial systems to greater security risks, including intrusion attempts and cyberattacks. The multitude of sensors present in these networks generates a considerable amount of data, attracting the attention of cybercriminals worldwide. To protect Industrial Internet of Things networks, applications against these attacks and intrusion detection systems play a crucial role. However, they have limitations in terms of detection accuracy and false alert management. By using machine learning and deep learning techniques, it is possible to mitigate the multiple security threats and enhance the ability of intrusion detection systems to identify complex attacks patterns and to adapt configurations against new threats. Indeed, such artificial intelligence methods can analyze vast amounts of real-time data, detect anomalies, identify known attack signatures and even predict potential attacks. This article constitutes a bibliographical overview of artificial intelligence based intrusion detection approaches as well as the different datasets on which they have been tested. Additionally, it aims to identify current limitations and challenges in ongoing and existing researches and solutions, while providing some directions for further scientific works.

Keywords—Artificial intelligence, Cybersecurity, IIoT, Intrusion detection

I. INTRODUCTION

The Industrial Internet of Things (IIoT) is Modernizing the way industries operate and transforming traditional manufacturing processes into intelligent and interconnected systems. Through the integration of advanced sensors, devices and data analytics, the IIoT enables seamless connectivity and real-time communication between machines, systems, and humans in industrial environments. By harnessing the power of the Internet and cloud computing, the IIoT allows organizations to gather vast amounts of data from various sources, such as sensors, machines and production lines. These data are then analyzed and utilized to optimize operations, improve productivity and make informed decisions. As the IIoT continues to evolve, it brings forth new opportunities and challenges. Data security [1] and privacy concerns are among the key considerations to address.

Especially, intrusion detection in IIoT systems play crucial role aiming to protect networks and connected devices used in industrial environments. With the growth of IIoT, ensuring the security of these networks is essential. Intrusion detection involves network traffic monitoring, behavioral analysis, the use of known attack signatures and device-level security integration. The objective is to identify suspicious activities and take measures to counter them. This field is constantly evolving to address emerging threats. Intrusion detection holds decisive importance in IIoT systems due to the numerous security challenges they face. In fact, IIoT systems are exposed to a wide range of threats, Such as denial-of-service attacks, data breaches, and compromised confidentiality. The distributed and interconnected nature of these systems, combined with the diversity of devices and communication protocols used, creates specific challenges in ensuring their security.

Intrusion detection plays a vital role by continuously monitoring network traffic, analyzing IoT device behaviors, and identifying suspicious activities. This enables organizations to react promptly to security incidents, mitigate potential damages, and enhance the resilience of their industrial IoT systems [2]. In this context, Artificial intelligence (AI) plays essential role in intrusion detection in IIoT systems, enhancing security and improving the ability to detect potential attacks.

In this article, we attempt to highlight various AI-based intrusion detection approaches and the datasets used to test them. It also aims to identify current limitations as well as some questions that could be the subject of future scientific researches in order to bring solutions and to improve these systems.

II. ROLE OF AI IN INTRUSION DETECTION IN IIOTs

First, In the context of IIoT, the role of AI can be summarized in several key aspects. Firstly, AI enables advanced behavior analysis by developing machine learning models capable of analyzing and understanding the normal behaviors of IIoT devices. These models can quickly identify anomalies or suspicious behaviors, such as subtle changes in traffic patterns or unexpected activities, indicating a possible intrusion. Secondly, AI contributes to the detection of unknown attacks by learning patterns and recognizing abnormal behavioral patterns, even in the absence of predefined signatures. Unlike tra-

ditional intrusion detection systems that rely solely on known attack signatures, AI-based approaches can identify emerging or unknown attacks by continuously learning and adapting to evolving threats. Furthermore, AI helps reduce false positives, which is a major challenge in intrusion detection, by leveraging machine learning algorithms to enhance detection accuracy. By analyzing and understanding traffic patterns and behaviors, AI can filter out unnecessary alarms and focus on true intrusion events. Moreover, AI ensures adaptability to new threats by continuously learning and updating detection models. As new attack techniques emerge, AI algorithms can adjust and improve the detection capabilities of IIoT systems. Finally, AI enables the automation of incident response by evaluating and categorizing intrusions based on severity, triggering appropriate mitigation measures. This facilitates swift and efficient response to attacks, minimizing potential damages. In conclusion, the integration of AI in intrusion detection systems for IIoT enhances detection accuracy, adaptability to emerging threats, and the ability to automate incident response. However, it is crucial to ensure data quality and rigorous model training to ensure the reliability and effectiveness of AI-based intrusion detection systems [3].

A. AI-based approaches for IIoT intrusion detection

Intrusion detection in IIoT systems benefits from various artificial intelligence (AI) approaches, including machine learning and deep learning methods as well as anomaly detection techniques. In this section, we present an overview of these approaches.

a) *Machine Learning*: Machine learning is a branch of AI that enables systems to detect patterns and anomalies from data without being explicitly programmed. In the context of intrusion detection, machine learning algorithms can be trained on historical data to recognize normal behaviors and detect abnormal behaviors that may indicate an intrusion. They can also be used to create predictive models to anticipate potential attacks [4], [5].

b) *Deep Learning*: Deep learning is a subset of machine learning that employs artificial neural networks containing multiple layers to acquire layered understandings of data. In our context, deep neural networks are used to automatically extract complex features from network traffic data, system logs, and other relevant data sources. This approach can enhance the ability to detect abnormal behaviors and improve the accuracy of intrusion detection [6].

c) *Anomaly Detection Techniques*: Anomaly detection techniques are used to identify behaviors that deviate from the norm. These techniques can be applied to detect suspicious and/or malicious activities in IIoT systems. Anomaly-based approaches can employ statistical methods, machine learning algorithms, or clustering techniques to detect abnormal behaviors and potential intrusions [7].

B. AI-based approaches for IIoT intrusion detection

The above AI approaches offer advanced capabilities for intrusion detection in industrial IoT. The use of these approaches

in combination with appropriate datasets and effective feature engineering can improve the accuracy and efficiency of intrusion detection in IIoT systems. Indeed, evaluation datasets are important to validate any intrusion detection approach. It allows us to assess the proposed method's ability to detect intrusive behavior.

There are several publicly available datasets that are widely used by researchers and developers as benchmarks in this field. Among them, we cite Bot-IoT, MQTT-IoT-IDS2020, IoT-23, TON-IoT, UNSW-NB 15, CICIDS2017 and NSL-KDD dataset.

- The Bot-IoT dataset is used for intrusion detection in IoT systems. It has been specifically designed to simulate botnet attacks targeting IoT devices. The dataset includes network traffic records generated by botnet attacks in realistic scenarios. It provides a variety of malicious behaviors and attack patterns, making it a valuable resource for evaluating and enhancing intrusion detection techniques in IoT systems. The Bot-IoT dataset can be downloaded online [8], [9].
- The MQTT-IoT-IDS dataset is a dataset commonly used for intrusion detection in MQTT-based IoT systems. It consists of network traffic data collected from MQTT (Message Queuing Telemetry Transport) communication protocol used in IoT environments. The dataset contains both normal and malicious MQTT traffic, including various types of attacks and anomalies. It serves as a valuable resource for developing and evaluating intrusion detection systems specifically tailored to MQTT-based IoT architectures [10].
- IoT-23 dataset has been specifically designed for intrusion detection in IoT systems. It includes a variety of IoT devices such as Somfy door lock, Philips Hue and Amazon Echo devices. The data is collected by simulating realistic attack scenarios, enabling the evaluation of intrusion detection algorithm performance in industrial IoT environments. The aim is to provide researchers with an extensive dataset that includes authentic and labeled instances of IoT malware infections, as well as benign IoT traffic [11].
- The ToN-IoT dataset is created to assess blended data from IoT and industrial IoT, including diverse sources like device telemetry, system logs of both Windows and Linux platforms, as well as network traffic within the system. The dataset aims to provide a comprehensive view of data generated in IoT and industrial IoT environments, enabling researchers to develop and evaluate intrusion detection techniques that can handle diverse data types and sources [12], [13].
- The UNSW-NB15 dataset, created at UNSW Canberra's Cyber Range Lab, comprises a mix of real modern network activities and synthetic contemporary attacks. It includes nine attack types such as Backdoors, Fuzzers, Analysis, DoS, Exploits, Generic, Shellcode, Reconnaissance, and Worms. [14], [15].

- The CICIDS2017 dataset contains common and recent attacks, as well as realistic network traffic resembling real-world data. It includes results from network traffic analysis and labeled flows. A system called B-Profile was used to generate natural and benign background traffic based on the behavior of 25 users [16], [17].
- The NSL-KDD dataset is an enhanced version of the KDD Cup 1999 dataset used for intrusion detection in networks. It addresses the limitations of the previous dataset by eliminating redundant and similar data. It includes both normal and abnormal network connections, covering various attacks such as denial of service, remote-to-local, user-to-root, and probing. It is widely used for developing machine learning-based intrusion detection algorithms [18], [19].

The table 1 below presents a summary of IDS Datasets generated from the real network traffic traces

III. MODELS FOR IIoT WITH LIMITATIONS

Before Several works have been done in the field of constructing intrusion detection systems for the IIoT, leveraging a variety of machine learning and deep learning models. These advancements aim to enhance the security of IIoT applications by identifying and preventing potential intrusions in connected industrial networks. In [20] Mengmeng, Ge. et al. have employed a Feedforward neural network (FNN) model to create an intelligent binary and multiclass classification scheme. They have extracted and preprocessed information from individual packets to use as generic features. The effectiveness of this scheme was evaluated using the Bot-IoT dataset. The classifier has demonstrated a high capacity for binary classification, achieving nearly 99% precision, recall, and F1 score for DDoS/DOS attacks and reconnaissance attacks. For multiclass classification, their classifier has achieved a detection precision of over 99% for DDoS/DOS attacks, while normal traffic classification has been achieved with a precision of 98%. Abdalgawad, N., et al. [21] have used generative deep learning techniques, specifically Bidirectional Generative Adversarial Networks (BiGAN) and Adversarial Autoencoders (AAE), for intrusion detection by analyzing network data. They have employed the IoT-23 dataset for their experiments. The AAE and BiGAN models have both achieved an F1 score of 99%. Additionally, a BiGAN model have been trained to detect unknown attacks, including zero-day attacks, with an F1 score ranging between 85% and 100%. In [22], Roopak, M. et al., have proposed and evaluated deep learning models for DDoS attack detection using the CICIDS2017 datasets. They have implemented four different deep learning models and compared them to machine learning algorithms. They have found that the hybrid Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) model outperformed the other deep learning models and machine learning algorithms, achieving an accuracy of 97.16%. On the other hand, the multilayer perceptron deep learning model have showed the lowest performance on the used dataset. The three other deep

learning methods have all achieved an accuracy higher than 95.00%, demonstrating their superior performance compared to the machine learning algorithms. Bipraneel, R. et al., [23] have employed a deep learning-based method to detect attacks in IoT networks. They have utilized a recurrent neural network (RNN) Bidirectional Long Short-Term Memory (BLSTM) to train a multi-layer deep neural network. The training was conducted using the UNSW-NB15 dataset. The focus have been on the binary classification of normal and attack patterns in the IoT network. The proposed BLSTM RNN model have achieved an accuracy of over 95% in attack detection. Derhab, A., et al. [24] have used five design principles to develop a deep learning-based intrusion detection system was developed for IoT. It incorporates a Temporal Convolutional Neural Network (TCNN), a fusion of Convolutional Neural Network (CNN) and causal convolution. TCNN was evaluated on the Bot-IoT dataset and compared to methods such as logistic regression, random forest, LSTM, and CNN. The results indicate that TCNN achieves an accuracy of 99.998% for multiclass traffic detection. Elike, H. et al. [25] have adopted a neural network-based approach for intrusion detection in IoT networks, with a focus on identifying DDoS/DOS attacks. Their method has relied on classifying normal and threat patterns using an artificial neural network model. To validate their model, they have tested it on a simulated IoT network and has achieved an accuracy of over 99%. Their model has successfully identified different types of attacks and has demonstrated good performance in terms of true and false positive rates. Awotunde, J. et al., [26] have proposed an Anomaly Detection System (ADS) model to identify destructive activities in IIoT networks using data from TCP/IP packets. They have utilized unsupervised learning strategies based on deep learning that combine hybrid rule-based approaches with automated dimensionality reduction. This approach has effectively described standard network structures for unsupervised learning. They have employed a DAE-DFFNN model with a hybrid rule-based design. Compared to other research strategies, their model has achieved a maximum identification rate of 99.0% when tested on different data samples from the NSL-KDD and NSW-NB15 datasets. Naveen, S. et al., [27] have introduced an Intrusion Detection System using multiple machine learning classification techniques on the MQTT-IoT-IDS2020 dataset to identify multi-class intrusion attacks in the IoT. A comparative analysis of the results has also been conducted to evaluate the performance of all implemented machine learning classification techniques by the authors. The experiments have resulted in an overall accuracy of over 97% for all implemented classifiers including k-nearest neighbors, support vector machines, naive Bayes, random forest, decision tree and stochastic gradient descent.

The next table 2 presents a summary of related work in the field of intrusion detection in IoT networks, as discussed earlier. The various approaches and methods are presented, along with their respective authors, descriptions of the used approaches, evaluated datasets, and achieved performances.

TABLE I: SUMMARY OF IDS DATASETS GENERATED FROM THE REAL NETWORK TRAFFIC TRACES

Dataset	Year	Developed By	Publicly available	Attack
Bot-IoT	2019	The Cyber Range Lab of UNSW Canberra	Yes	DDoS, DoS, Keylogging, Data exfiltration, Reconnaissance
MQTT-IoT-IDS	2020	Hanan Hindy (Abertay University)	Yes	aggressive scan, UDP scan, Sparta SSH, brute-force, MQTT brute-force attack
IoT-23	2019	The Stratosphere Laboratory, AIC group, FEL, CTU University	Yes	DDoS, FileDownload, HeartBeat, Okiru, C2, PortScan
ToN-IoT	2021	The Cyber Range Lab of UNSW Canberra	Yes	DoS, DDoS, Ransomware, Backdoor, MITM, password violations, Xss, Scanning, injection
UNSW-NB15	2015	The Cyber Range Lab of UNSW Canberra	Yes	Analysis, Backdoors, DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode, Worms
CICIDS2017	2017	Canadian Institute of Cyber Security	Yes	Botnet, Brute Force, DDoS, DoS, Heartbleed, Infiltration, Port Scan, Web
NSL-KDD	2009	University of California	Yes	NormalDoS, Remote-to-local, User-to-root, Probing

TABLE II: SUMMARY ON INTRUSION DETECTION IN IOT APPROACHES USING IA

Authors	Approach	Year	Dataset	Performance
Mengmeng, Ge. et al. [20]	FNN neural network model to create an intelligent binary and multiclass classification scheme	2019	BoT-IoT	Recall of 99%, Accuracy of 98%
Abdalgawad, N., et al. [21]	Generative deep learning methods to detect intrusions by analyzing network data	2021	IoT-23	F1 score of 99%
Roopak, M. et al. [22]	Evaluated deep learning models for DDoS attack detection	2019	CICIDS2017	Accuracy of 97.16%
Bipraneel, R. et al. [23]	A method based on deep learning, employing a recurrent neural network (RNN) bidirectional Long Short-Term Memory (BLSTM)	2018	UNSW-NB15	Accuracy of over 95%
Derhab, A., et al. [24]	Five design principles to develop a deep learning-based intrusion detection system for the Internet of Things (IoT)	2020	Bot-IoT	Accuracy of 99.998%
Elike, H. et al. [25]	Neural network-based approach for intrusion detection in IoT networks, with a focus on identifying DDoS/DOS attacks	2016	Data from various parts of the IoT network	Accuracy of over 99%
Awotunde, J. et al. [26]	Anomaly Detection System (ADS) model to identify destructive activities in IIoT networks	2021	NSL-KDD and NSW-NB15	Accuracy of 99.0%
Naveen, S. et al. [27]	Intrusion Detection System using multiple machine learning classification techniques	2023	MQTT-IoT-IDS2020	Accuracy of over 97%

IV. CHALLENGES AND LIMITATIONS

As all systems, intrusion detection on IIoT devices presents challenges and limitations that need to be considered to ensure effective security of IIoT. In this section, we will explore the major challenges faced by intrusion detection on IIoT using AI, as well as the inherent limitations of the AI approaches. Namely, data preprocessing, networks and systems complexity as well as scalability and performance, are the major challenges and limitations that should be understood in order to develop more robust solutions and also to enhance the security of IIoT against persistent threats.

- **Data preprocessing:** Data from IIoT sensors can be complex, noisy, and unstructured. They can also exhibit class imbalances in intrusion detection, making data preprocessing crucial to ensure the quality of input data for the deep learning model.
- **Complexity of the network and systems:** IIoT systems can be extremely complex, with numerous devices, sensors, and different communication protocols. Designing an IIoT intrusion detection system must take this complexity into account to effectively detect potential intrusions while minimizing false positives and false negatives.
- **Scalability and performance:** IIoT systems often generate real-time data streams which require algorithms capable of efficiently and timely processing the data. System scalability must be taken into account to handle large-scale deployments and frequent model updates.

The figure 1 below provides an overview of the challenges and limitations that require constant attention to ensure their effectiveness and relevance in intrusion detection in the IIoT.

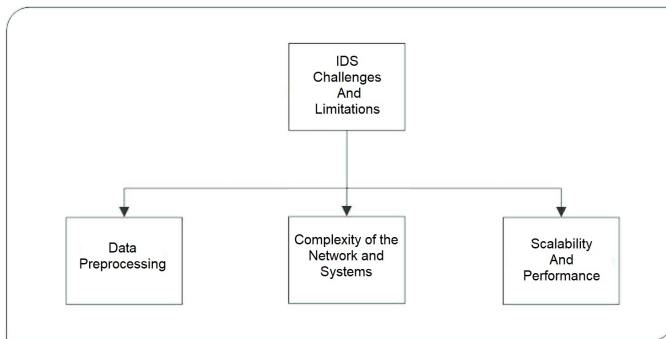


Fig. 1: Intrusion Detection System Challenges And Limitations

V. CONCLUSION AND FUTURE DIRECTIONS

In conclusion, this article highlights the growing importance of intrusion detection in Industrial Internet of Things (IIoT) systems and the essential role that artificial intelligence can play in this field, along with the most commonly used datasets by researchers. Indeed, IIoT offers significant advantages in terms of operational efficiency and productivity, but it is also vulnerable to attacks and intrusions. AI-based approaches, such as machine learning and neural networks, have demonstrated their ability to detect malicious activities and enhance

the security of IIoT. However, there are still several challenges to overcome. The main of them remains the complexity of the IIoT network and systems. Furthermore, attacks on IIoT are constantly evolving, requiring AI techniques capable of detecting new forms of intrusion and adapting rapidly.

In our future work, we plan to use multiple machine and deep learning models to apply them on the TON-IIoT dataset. This dataset is the most recent and least utilized by researchers. It provides heterogeneous data from different types of IIoT systems, communication protocols, and sensors. This will allow us to capture a variety of potential intrusion behaviors and scenarios, making the conceived model more robust and generalizable.

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